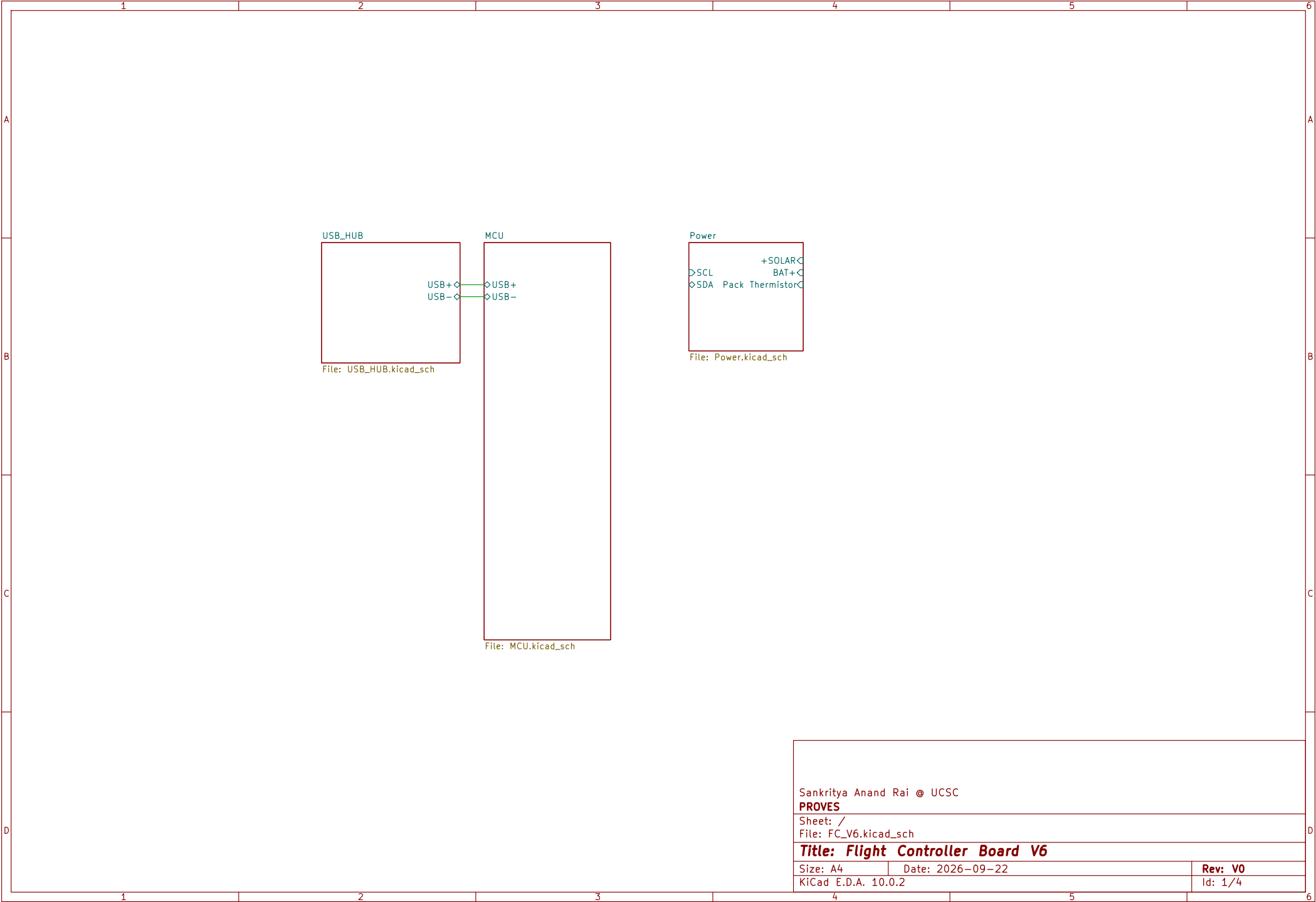




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PROVES

Sheet: /

File: FC_V6.kicad_sch

Title: Flight Controller Board V6

Size: A4

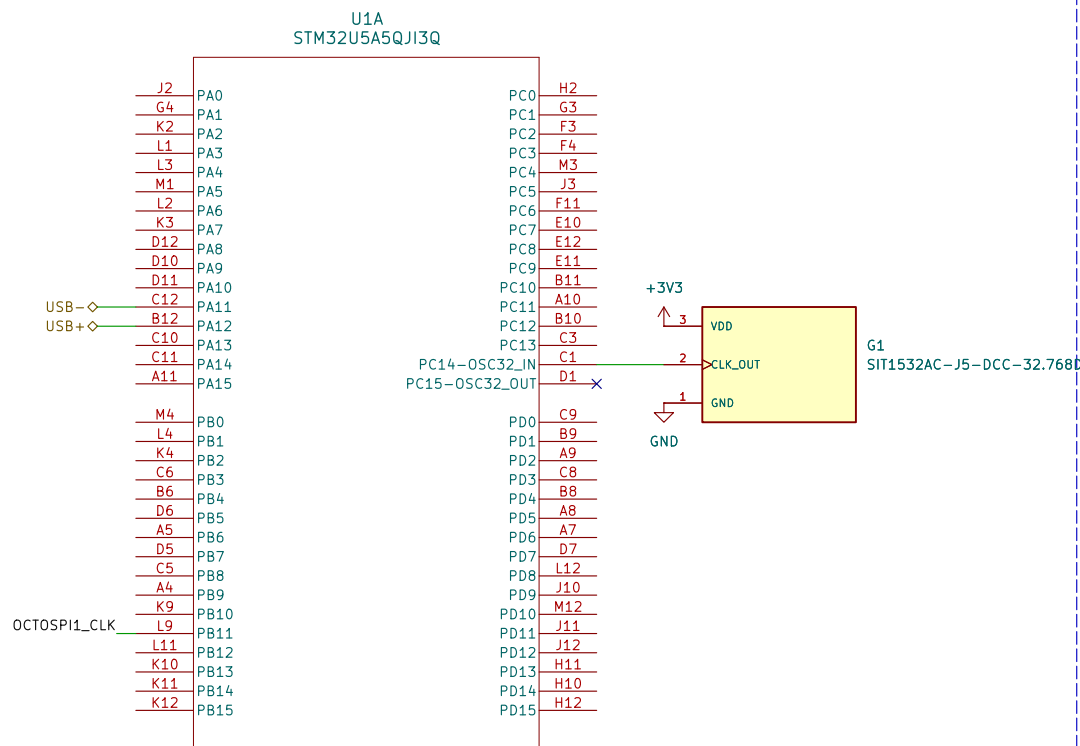
Date: 2026-09-22

Rev: V0

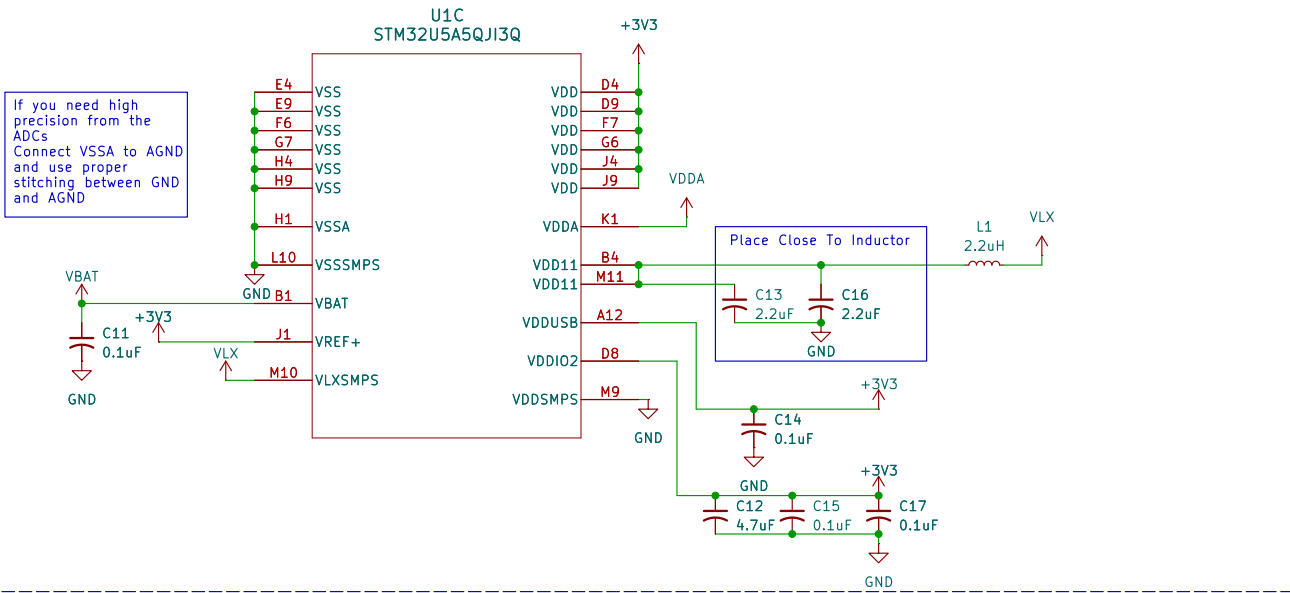
KiCad E.D.A. 10.0.2

Id: 1/4

GPIO



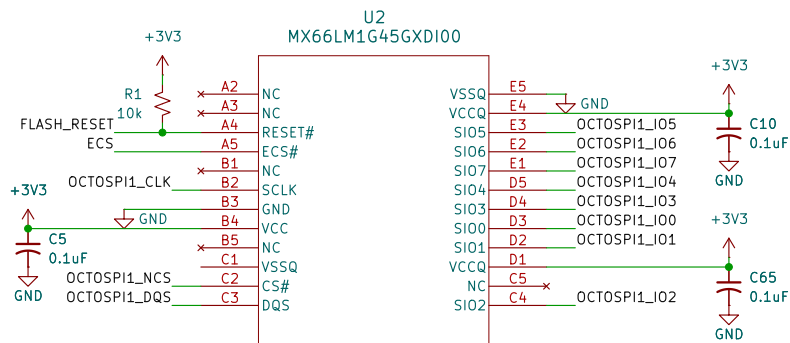
Power



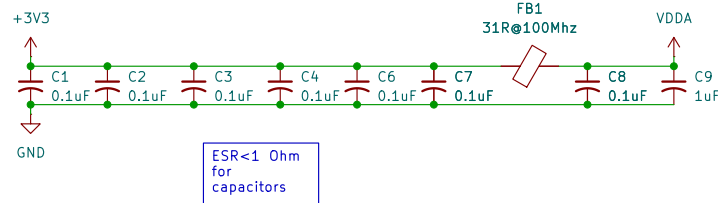
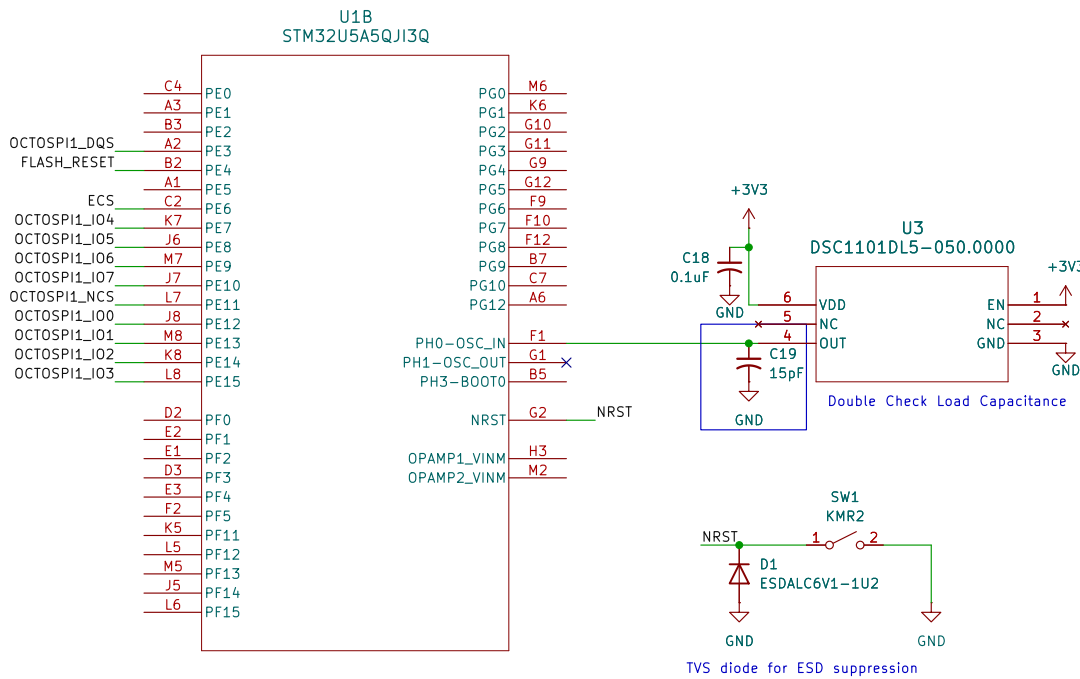
OCTOSPI Flash

Keeping the data lines (IO0-IO7) on the same physical port (Port E) ensures the internal silicon routing and GPIO toggle timings are as closely matched as possible. Route DQS exactly like a data line. Match the lengths of IO[0:7], CLK, and DQS to within ±2mm of each other to prevent timing skew during high-frequency operations. Ensure all 11 signals are routed over a continuous, unbroken ground plane on the adjacent layer to provide a clean return path and maintain the target single-ended impedance (typically 50Ω).

ECS pin will report an error corrected signal, used to log flash degradation. Reset should generally not be externally toggled by the MCU but in the event of a hard reset or crash it can prove useful.

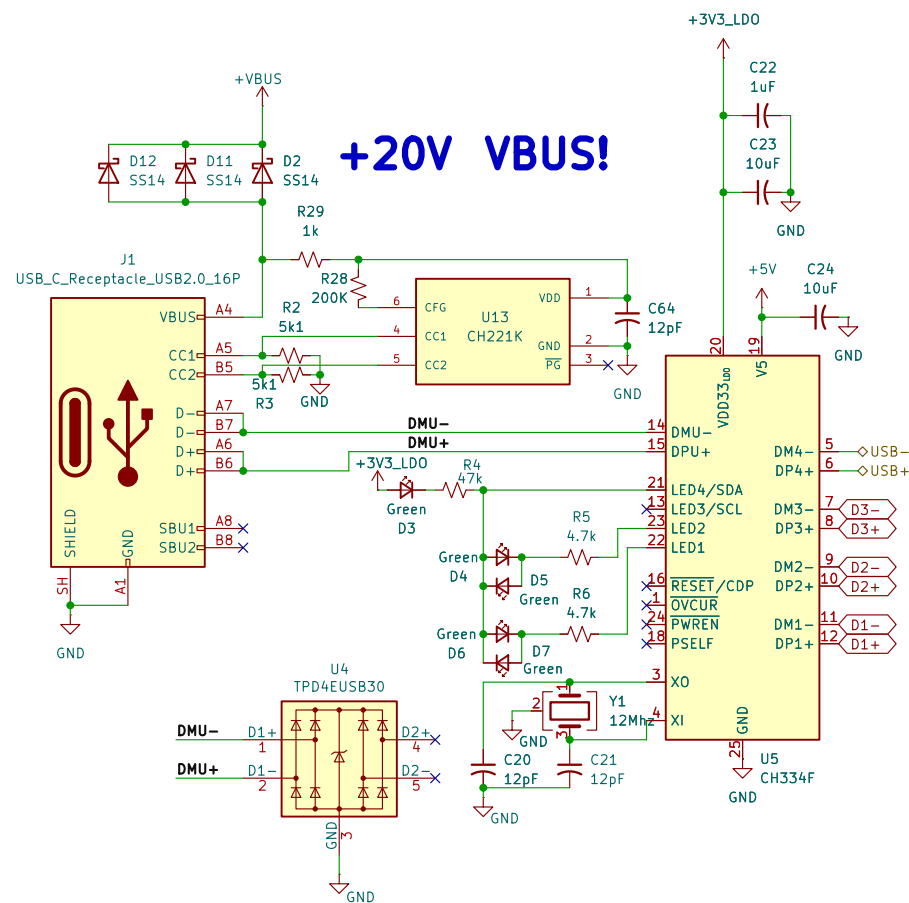


GPIO + CLK + RESET



Power Supply Decoupling

Under No circumstances should the LEDs or Load Capacitors be made DNP,
 Doing so breaks the USB enumeration process on Windows 11 and stops the hub working entirely specifically on dell laptops,
 Donot believe the datasheet
 This setup has been validated and works.



Standard USB 2.0 Spec routing

Sheet: /USB_HUB/
 File: USB_HUB.kicad_sch

Title: USB Hub + PD

Size: A4 Date: 2026-09-23
 KiCad E.D.A. 10.0.2

Rev:
 Id: 3/4

